

Data Flow

Create a Data Flow Job Using Template

1. Open Google Cloud Console and navigate to Dataflow.
2. Click Create Job from Template.
3. Enter Job Name: df-demo.
4. Select Region: us-central1.
5. Choose the predefined template: Text Files on Cloud Storage to BigQuery using JavaScript UDF.
6. Create a Cloud Storage bucket and upload the required files: employees_ex.csv, udf.js, and bq.json.
7. Create an empty BigQuery Dataset named df_exdemo.
8. Create an empty BigQuery table named employees with the schema from bq.json.
9. In the Dataflow template, provide the input file path from GCS.
10. Provide the JavaScript UDF path (udf.js) stored in GCS.
11. Provide the BigQuery output table path project.dataset.table.
12. Provide the service account with Dataflow, BigQuery, and Storage permissions in Optional Parameters.
13. Review all fields and click Run Job.
14. Monitor the job execution until all stages show Succeeded.
15. Verify that the records are loaded into BigQuery.
16. Run a SELECT * query on the employees table to validate the output.

Screenshots

Files uploaded successfully to GCS bucket.

← Bucket details

Edit bucketGo to pathRefresh

df-demo-007

Location

Storage class

Public access

Protection

us (multiple regions in the United States)

Standard

Not public

Soft delete

Objects

Configuration

Permissions

Protection

Lifecycle

Observability

Inventory Reports

Operations

Folder browser

df-demo-007

Buckets > df-demo-007

Create folderUploadTransfer data

Filter by name prefix onlyFilter objects and foldersShowLive objects only

	Name	Size	Type	
	bq.json	409 B	application/json	⬇ ⋮
	employees_ex.csv	1.8 KB	text/csv	⬇ ⋮
	udf.js	461 B	text/javascript	⬇ ⋮

BigQuery dataset and table created.

Google CloudMy First Project

Search (/) for resources, docs, products and more

Search

Search BigQuery resources

Show starred only

project-e1867c45-6603-4a6f-9...

- Queries
- Notebooks
- Data canvases
- Data preparations
- Pipelines
- Connections
- df_exdemo
 - employees

Create new

What's new in Studio

SQL queryNotebookNotebook with SparkML modelGraphAgentConversationPipelineData preparationData canvasTable

Recent

Display name	Type	Last modified time	Project
employees	Table	17 Jun 2026, 14:25:45	project-e1867c45-6603-4a6f-949
df_exdemo	Dataset	17 Jun 2026, 13:49:46	project-e1867c45-6603-4a6f-949

Try with templatesView notebook gallery

Add your own data

Dataflow job execution details.

df-demo Clone Stop Archive Send feedback

Job graph Execution details Job metrics Cost Recommendations

Job steps view
Graph view Clear selection

Read from source
Succeeded
3 sec
2 of 2 stages succeeded

Javascript...avascript
Succeeded
1 sec
1 of 1 stage succeeded

Logs Show

Job info

Job name	df-demo
Job ID	2026-06-17_01_58_16-6281405926164837070
Job type	Batch
Job status	Succeeded
SDK version	Apache Beam SDK for Java 2.74.0
Job region	us-central1
Current workers	0
Latest worker status	Worker pool stopped.
Start time	17 June 2026, 14:28:17 GMT+5
Elapsed time	4 min 46 sec

Dataflow BigQuery write stage succeeded.

df-demo Clone Stop Archive Send feedback

Job graph Execution details Job metrics Cost Recommendations

Job steps view
Graph view Clear selection

BigQueryC...oTableRow
Succeeded
0 sec
1 of 1 stage succeeded

Insert in... Bigquery
Succeeded
7 sec
20 of 20 stages succeeded

Logs Show

Job info

Current workers	0
Latest worker status	Worker pool stopped.
Start time	17 June 2026, 14:28:17 GMT+5
Elapsed time	4 min 46 sec
Encryption type	Google-managed
Dataflow Prime	Disabled
Knowledge Catalog lineage	Disabled
Runner v2	Enabled
Dataflow Shuffle	Enabled

Resource metrics

Current vCPUs	1
---------------	---

BigQuery table preview with loaded records.

Row	emp_id	emp_name	department	salary	city	emp_email
1	1003	ARJUN MEHTA	Finance	62000	Bangalore	arju...
2	1008	MEENA PATEL	Finance	68000	Ahmedabad	mex...
3	1012	DIVYA NAIR	Finance	61000	Kochi	divy...
4	1017	AMIT TIWARI	Finance	64000	Lucknow	ami...
5	1022	SHALINI BOSE	Finance	66000	Kolkata	sha...
6	1002	SNEHA REDDY	HR	48000	Chennai	sne...
7	1006	ANJALI VERMA	HR	51000	Delhi	anj...
8	1013	MANOJ GUPTA	HR	49000	Delhi	mar...
9	1018	KAVYA MENON	HR	50000	Kochi	kav...
10	1023	VENU GOPAL	HR	52000	Hyderabad	ven...
11	1001	RAVI KUMAR	IT	55000	Hyderabad	ravi...

Query results validating loaded data.

Query completed
Using on-demand processing quota

Row	emp_id	emp_name	department	salary
1	1003	ARJUN MEHTA	Finance	
2	1008	MEENA PATEL	Finance	
3	1012	DIVYA NAIR	Finance	
4	1017	AMIT TIWARI	Finance	
5	1022	SHALINI BOSE	Finance	
6	1002	SNEHA REDDY	HR	

Result

The Dataflow template job completed successfully. The CSV data was transformed using the JavaScript UDF and loaded into the BigQuery table df_exdemo.employees. The final output was validated using BigQuery Preview and SQL query results.

Code

1.BQ.json

Code

1.BQ.json

```
{
  "BigQuery Schema": [
    { "name": "emp_id", "type": "INTEGER" },
    { "name": "emp_name", "type": "STRING" },
    { "name": "department", "type": "STRING" },
    { "name": "salary", "type": "INTEGER" },
    { "name": "city", "type": "STRING" },
    { "name": "email", "type": "STRING" },
    { "name": "joining_date", "type": "DATE" },
    { "name": "annual_salary", "type": "INTEGER" }
  ]
}
```

2.UDF.js

```
function process(line) {

  var values = line.split(",");

  if (values[0] === "emp_id") {
    return null;
  }

  var obj = {};

  obj.emp_id = parseInt(values[0], 10);
  obj.emp_name = values[1].toUpperCase();
  obj.department = values[2];
  obj.salary = parseInt(values[3], 10);
  obj.city = values[4];
  obj.email = values[5];
  obj.joining_date = values[6].trim();
  obj.annual_salary = obj.salary * 12;

  return JSON.stringify(obj);
}
```